INSTITUTE OF ENGINEERING & MANAGEMENT (IEM)
Salt Lake, Kolkata
Department of Computer Science & Engineering

PCCCS504 | Software Engineering and Project Management | Semester V | L:T:P:S – 3:0:3:0

SOFTWARE REQUIREMENTS SPECIFICATION

Lab 3 – Requirements Engineering | Assignment 3 | IEEE 830 structure

Student / Team Details

Team No.:	P4
Team Name:	DevForge
Section:	B
Batch:	2024-28
Semester / AY:	5th
Faculty In-Charge:	Saikat Dutta , Partha Chakraborty
Date:	19/08/2026

Team Members

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Purpose of This Document

This document lists everything your system must do. It is used again in Lab 4 (UML), Lab 6 (testing), and Lab 7 (final evaluation), so fill it in carefully.

Follow the structure below in order. You need at least 5 Functional Requirements and 3 Non-Functional Requirements. Each one should be clear and testable — avoid vague lines like ‘the system should be fast’.

1. Introduction

1.1 Purpose

Purpose of this document
The purpose of this document is to specify the functional and non-functional requirements of the IEM Admission System. It is intended to guide the project team through design, development, and testing, and to serve as a reference for the faculty in-charge and other stakeholders in evaluating whether the delivered system meets its intended requirements.

1.2 Scope

Scope of the product being specified
We will build an online college admission application to streamline student registration, form submissions, document uploads, and staff verifications for IEM. It allows prospective students to register themselves, complete and submit admission forms, and upload the necessary documents digitally. At the same time, authorized admission staff can verify applications, maintain applicant records, and provide updates regarding admission status through a secure system.

1.3 Definitions, Acronyms & Abbreviations

Term	Definition
FR	Functional Requirement — a specific behaviour the system must perform.
NFR	Non-Functional Requirement — a quality attribute the system must satisfy.
IF	Interface Requirement — a boundary through which the system exchanges data with users or other systems.

1.4 References

References (textbooks, standards, tools used)
PCCCS504 – Software Engineering and Project Management, Lab 3 course material, Dept. of CSE, IEM Kolkata.
Project Charter — IEM Admission System (team-authored, August 2026).
IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

1.5 Overview

Overview of the rest of this document
The remainder of this document is organized as follows. Section 2 gives an overall description of the product, including its perspective, main functions, intended users, constraints, and assumptions. Section 3 lists the detailed functional requirements. Section 4 lists the non-functional requirements. Section 5 lists the interface requirements. Section 6 contains appendices for supporting material.

2. Overall Description

2.1 Product Perspective

Product Perspective

The IEM Admission System is a new, self-contained web application built to replace manual, paper-based admission handling at IEM. It consists of a student-facing admission portal, an administrator dashboard for admission staff, a shared database that stores applicant and application data, an online payment interface for admission fees, and an email/SMS notification interface for status updates. It does not depend on any existing legacy admission software.

2.2 Product Functions

Note: A short list of the main functions. The full detailed list goes in Section 3.

Product Functions (summary)

At a summary level, the system will allow students to register, complete and submit admission applications, upload documents, pay fees online, and track application status; and will allow administrators to view, verify, approve/reject applications, request corrections, search/filter applications, and generate reports. The detailed list of functions is provided in Section 3.

2.3 User Characteristics

User Characteristics

There are two primary classes of users. Applicants are prospective students, typically first-time users of the system with varying levels of computer literacy, who require a simple and guided interface to complete their application. Administrators are IEM admission staff who are trained, regular users of the system responsible for verifying applications and require efficient tools for searching, filtering, and processing applications in bulk during peak admission periods.

2.4 Constraints

Constraints

- The project must be completed within a limited development period (03/08/2026 to 27/11/2026).
- Real admission records will not be available for development or testing.
- Sample/synthetic datasets will be used in place of real applicant data for all testing.
- The system must be built using tools and environments available to the team throughout the project.

2.5 Assumptions and Dependencies

Assumptions and Dependencies

- Every team member will complete their assigned tasks on time.
- The required software and development environments will remain available for the duration of the project.
- Faculty requirements for the system will remain stable throughout the project.
- An external payment gateway will be available and reliable for integration and testing.
- An email/SMS service provider will be available for sending notifications.

3. Functional Requirements

Note: Minimum 5 required. Each row should describe ONE thing the system must do, written so it can be tested later ('the system shall…').

FR ID	Description	Priority (H/M/L)
FR-1	The system shall allow students to create a new admission account using their personal details.	H
FR-2	The system shall allow students to securely log in and log out.	H
FR-3	The system shall allow students to enter and update their personal information.	H
FR-4	The system shall allow students to enter their academic qualifications and marks.	H
FR-5	The system shall allow students to select their preferred course/program.	H
FR-6	The system shall allow students to upload required admission documents.	H
FR-7	The system shall validate uploaded documents for supported file formats and size.	M
FR-8	The system shall allow students to review their completed application before submission.	H
FR-9	The system shall allow students to submit the admission application online.	H
FR-10	The system shall generate a unique application number for each submitted application.	H
FR-11	The system shall allow students to pay the applicable admission/application fee online.	H
FR-12	The system shall generate and display a payment reciept after successful payment.	M
FR-13	The system shall allow students to track the current status of their applications.	H
FR-14	The system shall allow authorized administrators to view submitted applications.	H
FR-15	The system shall allow administrators to verify student information and uploaded documents.	H
FR-16	The system shall allow administrators to approve or reject applications.	H
FR-17	The system shall allow administrators to request corrections or additional documents from applicants.	M
FR-18	The system shall allow administrators to search and filter applications by application number, course, or status.	M
FR-19	The system shall send notifications to students regarding application status and important updates.	M
FR-20	The system shall generate admission reports for authorized administrators.	L

4. Non-Functional Requirements

Note: Minimum 3 required. Cover things like Performance, Security, or Usability, each with a measurable target where possible.

NFR ID	Category	Description

NFR-1	Performance	The system shall load major pages and process normal user requests within 5 seconds under normal operating conditions.
NFR-2	Security	The system shall prote student credential and personal information using secure authentication and appropriate access controls.
NFR-3	Scalability	The system shall support an increasing number of students,applications,and concurrent users during peak admission periodswithout significant degradation in performance.
NFR-4	Usability	The system shall provide a simple,clear, and user-friendly interface so applicants can complete the admission process with minimal assistane.
NFR-5	Reliability	The system shall maintain application and payment data accurately and prevent loss of submitted information during normal operation.

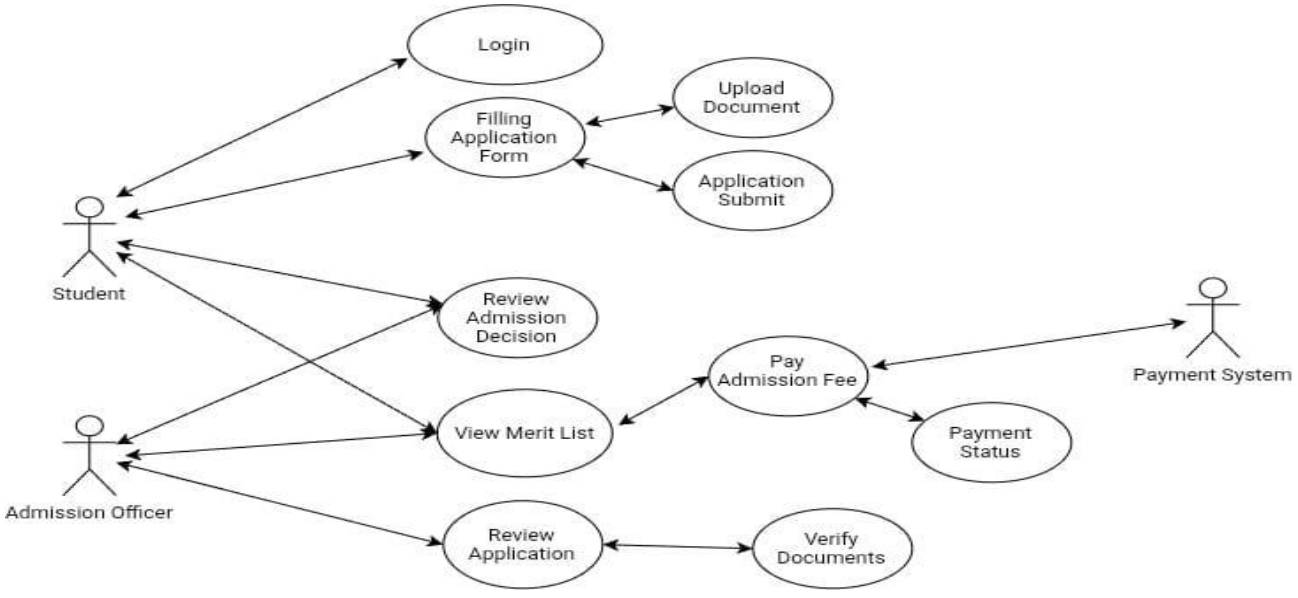
5. Interface Requirements

IR ID	Category	Description
IF-1	Student Admission Portal	Registration, Login, Application Form, Document Upload, Payment, Application Status
IF-2	Admission Staff Dashboard	Staff Login, View Applications, Verify Documents, Approve/Reject Applications
IF-3	Database Interface	Stores and retrieves student details, applications, documents, and payment records
IF-4	Online Payment Interface	Payment page connected to a payment gateway for application/admission fees
IF-5	Email/SMS Notification Interface	Sends registration confirmation, payment confirmation, approval/rejection, and status updates

6. Appendices

Appendix (supporting diagrams, glossary, open issues)
A: Use Case Diagram — Illustrates the interactions between students, administrators, and the admission system.
B: Data Flow Diagram (DFD) — Illustrates how student and admission-related data flows through the system.

A) Use Case Diagram:



B) Data Flow Diagram (DFD):

